STRATHMORE UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

DIABETES PREDICTION USING MACHINE LEARNING:

Application of complete data science concepts

BY

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112721

A Data Science Concepts Assignment Submitted in Partial

Fulfillment of the Requirements for the Award of the

Degree of Master of Science in Information Technology

NAIROBI, KENYA

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Table of Abbreviations

Abbreviation	Description
ADAP	Adaptive Learning Algorithm
ADA	American Diabetes Association
BMI	Body Mass Index
IDF	International Diabetes Federation
LMIC	Low- and Middle-Income Countries
NCD	Non-Communicable Disease
NIDDK	National Institute of Diabetes and Digestive and Kidney Diseases
SDG	Sustainable Development Goal
UHC	Universal Health Coverage
WHO	World Health Organization

1 Problem Framing

This section frames the core problem addressed in this study. It provides a clear definition of the problem, identifies the stakeholders involved, explains the business and policy relevance, and situates the problem within its operational context.

1.1 Problem Definition

Non-communicable diseases (NCDs) represent one of the most pressing public health challenges of the 21st century, accounting for 74% of global deaths annually (**?**). Among NCDs, diabetes mellitus stands out as one of the most prevalent and rapidly growing conditions. According to the International Diabetes Federation, approximately 537 million adults (aged 20-79 years) were living with diabetes in 2021, and this number is projected to rise to 783 million by 2045 (**?**). Diabetes mellitus is a group of metabolic disorders characterized by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both (**?**). The World Health Organization classifies diabetes into several types, with Type 2 diabetes accounting for approximately 90-95% of all cases (**?**). Left undetected and untreated, chronically elevated blood glucose levels progressively damage multiple organ systems, leading to severe microvascular and macrovascular complications including diabetic retinopathy, diabetic nephropathy, cardiovascular disease, stroke, and lower limb amputation (**??**). Despite the availability of established diagnostic criteria—including fasting plasma glucose, oral glucose tolerance test, and HbA1c measurements (**?**)—a substantial proportion of individuals with diabetes remain undiagnosed. The IDF estimates that globally, approximately 44.7% of adults with diabetes are undiagnosed (**?**). This diagnostic gap is particularly pronounced in low- and middle-income countries (LMICs), where limited healthcare infrastructure, shortage of trained personnel, and inadequate laboratory facilities present significant barriers to timely screening and diagnosis (**??**). Undiagnosed diabetes represents both a clinical failure and a public health crisis. Individuals with undiagnosed diabetes often present to health facilities only at advanced stages of disease progression, when irreversible complications have already developed (**??**). This delayed diag-

nosis not only compromises individual health outcomes but also increases healthcare costs due to expensive management of preventable complications (**??**). The challenge is further compounded in resource-constrained settings where point-of-care diagnostic equipment may be unavailable or unreliable, and where patient wait times for laboratory results can be prohibitively long (**??**). Consequently, there is an urgent need for accessible, cost-effective, and accurate methods for early diabetes detection that can be deployed at scale, particularly in primary care and community-based settings where most at-risk populations seek initial care. This study asks: **Can we build a machine learning model that accurately identifies whether a female patient aged 21 and above is at risk of diabetes, using only routine clinical measurements available at primary healthcare level?** The goal is to develop a binary classification model that predicts either diabetic (1) or non-diabetic (0), flagging high-risk individuals early to enable timely medical intervention before complications arise.

1.2 Stakeholders

The successful deployment and impact of a diabetes prediction model depends on the engagement of multiple stakeholders across the healthcare ecosystem. Table **??** identifies the key stakeholders and their respective roles.

Table 2: Key Stakeholders in the Diabetes Prediction Model

Stakeholder	Role and Interest
Patients	Primary beneficiaries of early and accurate diagnosis; earlier detection improves treatment outcomes and quality of life
Clinicians & Nurses	Use model predictions to prioritize, triage, and refer high-risk patients for confirmatory laboratory testing
Hospital Administrators	Allocate screening resources efficiently and measure the impact of preventive care programmes
Ministry of Health (Kenya)	Design and implement evidence-based national diabetes screening and prevention policies
Health Insurers	Risk-stratify insured populations to design targeted preventive health packages and reduce claims
Data Scientists & Researchers	Develop, validate, and improve the predictive models using clinical and population-level data

1.3 Business and Policy Relevance

Beyond the clinical imperative, this study addresses critical economic and policy priorities. The economic case for early diabetes detection is compelling: while treatment of late-stage complications consumes the majority of diabetes-related healthcare expenditure (**?**), evidence demonstrates that screening and early intervention programs are cost-effective, with returns well below established willingness-to-pay thresholds (**?**). In Kenya's context, where severe shortages of endocrinologists and laboratory capacity constrain diagnostic reach (**?**), a validated prediction model enables task-shifting to community health workers and nurses at primary care level (**?**)—directly supporting the government's Community Health Strategy (**?**) and Universal Health Coverage roadmap (**?**). This work aligns with Kenya's constitutional right to health (**?**), the Kenya Health Policy 2014-2030's NCD prevention priorities (**?**), and international commitments under SDG 3.4 and the WHO Global Action Plan for NCDs (**??**). Fur-

thermore, deployment of such a model generates valuable epidemiological intelligence for evidence-based resource allocation through the Kenya Health Information System (**?**), while offering measurable benefits to multiple stakeholders including the Ministry of Health (progress toward UHC targets), county health departments (devolved planning under Kenya's governance structure (**?**)), the National Hospital Insurance Fund (reduced high-cost claims), and development partners seeking scalable, low-cost interventions for NCD prevention.

1.4 Context

This problem is situated within a **clinical decision-support context**, specifically targeting primary healthcare facilities in Kenya and similar resource-constrained environments where specialist endocrinologists are scarce, laboratory infrastructure is limited, and patients often present at advanced stages of disease. The dataset employed in this study is the **Pima Indians Diabetes Database**, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and first analyzed by **?** using the ADAP (Adaptive Learning Algorithm) neural network to forecast diabetes onset in a high-risk population of Pima Indian women. The dataset comprises diagnostic measurements from 768 female patients aged 21 years and above, documented across eight predictor variables—number of pregnancies, plasma glucose concentration (2-hour oral glucose tolerance test), diastolic blood pressure, triceps skinfold thickness, 2-hour serum insulin, body mass index (BMI), diabetes pedigree function, and age—alongside a binary outcome variable indicating diabetes status (**?**). Although the dataset originates from a specific ethnic cohort in the United States, its constituent clinical variables (glucose, BMI, blood pressure, insulin) represent universal biomarkers routinely assessed in primary care settings globally, including Kenya, thereby conferring clinical generalizability despite population-specific origins. Importantly, the predictive model developed in this study is designed to *augment*—not replace—clinical judgment: it generates a risk probability score that flags high-risk individuals for confirmatory laboratory testing, maintaining a human-in-the-loop decision architecture that ensures responsible, ethically grounded, and clinically actionable deployment (**?**).

2 Data Understanding & Wrangling

2.1 Dataset Description

The Pima Indians Diabetes Database comprises 768 patient records with 9 variables. The descriptive statistics presented in Figure **??** and Table **??** provide a comprehensive overview of all variables, their data types, and distributional characteristics.

```
----------------------------------------
                            Type    Mean     Std    Min     Max
Pregnancies                int64    3.85    3.37   0.00   17.00
Glucose                    int64  120.89   31.97   0.00  199.00
BloodPressure              int64   69.11   19.36   0.00  122.00
SkinThickness              int64   20.54   15.95   0.00   99.00
Insulin                    int64   79.80  115.24   0.00  846.00
BMI                      float64   31.99    7.88   0.00   67.10
DiabetesPedigreeFunction float64    0.47    0.33   0.08    2.42
Age                        int64   33.24   11.76  21.00   81.00
Outcome                    int64    0.35    0.48   0.00    1.00
```

Figure 1: Raw Descriptive Statistics Output from Data Analysis

Table 3: Formatted Dataset Variables and Descriptive Statistics

Variable	Type	Mean	Std	Min	Max
Pregnancies	int64	3.85	3.37	0	17
Glucose	int64	120.89	31.97	0	199
BloodPressure	int64	69.11	19.36	0	122
SkinThickness	int64	20.54	15.95	0	99
Insulin	int64	79.80	115.24	0	846
BMI	float64	31.99	7.88	0.0	67.1
DiabetesPedigreeFunction	float64	0.47	0.33	0.08	2.42
Age	int64	33.24	11.76	21	81
Outcome	int64	0.35	0.48	0	1

2.2 Handling Missing Values

Although the dataset contains no explicit NaN values, zero values in certain clinical variables (Glucose, BloodPressure, SkinThickness, Insulin, and BMI) are biologically

implausible and represent missing data. Table **??** summarizes the extent of these implicit missing values.

```
----------------------------------------
Pregnancies                      0.0
Glucose                          0.0
BloodPressure                    0.0
SkinThickness                    0.0
Insulin                          0.0
BMI                              0.0
...
BloodPressure        :   35 zeros ( 4.56%)
SkinThickness        :  227 zeros (29.56%)
Insulin              :  374 zeros (48.70%)
BMI                  :   11 zeros ( 1.43%)
```

Figure 2: Raw descriptive missing values analysis

Table 4: Zero Values Analysis (Biologically Impossible Measurements)

Variable	Zero Count	Percentage
Glucose	5	0.65%
BloodPressure	35	4.56%
SkinThickness	227	29.56%
Insulin	374	48.70%
BMI	11	1.43%

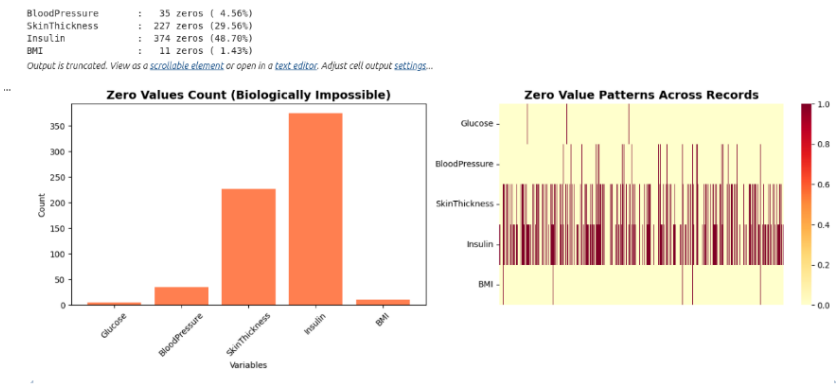


Figure 3: Truncated outputs after imputation

These missing values were imputed using the median value stratified by diabetes outcome (0 or 1), preserving the relationship between predictors and the target variable while avoiding introduction of bias.

```
 Replacing zero values with median (grouped by Outcome):
----------------------------------------
Glucose: Filled 5 missing values
BloodPressure: Filled 35 missing values
SkinThickness: Filled 227 missing values
Insulin: Filled 374 missing values
BMI: Filled 11 missing values

 After Handling - Missing Values:
----------------------------------------
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

All missing values handled!
```

Figure 4: Truncated outputs after imputation

2.3 Handling Outliers

Outliers were detected using two complementary methods: the Interquartile Range (IQR) method and the Z-score method ($|z| > 3$). Rather than removing outliers—which would reduce sample size—winsorization was applied, capping extreme values at the 1st and 99th percentiles. This approach retains all records while mitigating the influence of extreme observations.

```
 Outlier Detection using IQR Method:
----------------------------------------
Pregnancies              :   4 outliers
Glucose                  :   0 outliers
BloodPressure            :  14 outliers
SkinThickness            :  87 outliers
Insulin                  :  51 outliers
BMI                      :   8 outliers
DiabetesPedigreeFunction :  29 outliers
Age                      :   9 outliers
```

Figure 5: Detecting outliers using IQR

```
 Outlier Detection using Z-Score Method (|z| > 3):
----------------------------------------
Pregnancies              :   4 outliers
Glucose                  :   0 outliers
BloodPressure            :   8 outliers
SkinThickness            :   4 outliers
Insulin                  :  19 outliers
BMI                      :   5 outliers
DiabetesPedigreeFunction :  11 outliers
Age                      :   5 outliers
```

Figure 6: Detecting outliers using Z-score method

```
Pregnancies: Capped to [0.00, 13.00]
Glucose: Capped to [67.67, 196.00]
BloodPressure: Capped to [44.00, 106.00]
SkinThickness: Capped to [10.00, 51.33]
Insulin: Capped to [24.34, 519.90]
BMI: Capped to [19.50, 50.76]
DiabetesPedigreeFunction: Capped to [0.09, 1.70]
Age: Capped to [21.00, 67.00]
```

Figure 7: Handling outliers using winsorization

2.4 Handling Inconsistencies

Data integrity checks confirmed no duplicate records and verified that all values fall within clinically plausible ranges. All variables maintained appropriate data types, with integer types for count data (Pregnancies, Age) and float types for continuous measurements (BMI, DiabetesPedigreeFunction).

```
 Checking Logical Ranges:
----------------------------------------
Pregnancies         : OK
Glucose             : OK
BloodPressure       : OK
BMI                 : OK
Age                 : OK

 Checking for Duplicate Records:
----------------------------------------
Duplicate rows found: 0

 Data Type Consistency:
----------------------------------------
Pregnancies                   int64
Glucose                     float64
BloodPressure               float64
SkinThickness               float64
Insulin                     float64
BMI                         float64
...
Outcome                       int64
dtype: object

 Data inconsistency checks complete!
```

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

Figure 8: Handling Inconsistencies output

2.5 Feature Engineering

Four new features were engineered to capture clinical domain knowledge and enhance model performance:

1. **BMI_Category**: Classified BMI into WHO categories (Underweight, Normal, Overweight, Obese)
2. **Glucose_Category**: Classified fasting glucose per ADA guidelines (Normal, Prediabetes, Diabetic)
3. **Age_Group**: Segmented age into life stages (Young Adult <30, Middle Age 30-50, Senior >50)
4. **Insulin_Resistance_Index**: Interaction term (Glucose × BMI / 100) capturing metabolic syndrome risk

These engineered features enable the model to capture non-linear relationships and clinical thresholds that simple linear combinations of raw variables might miss.

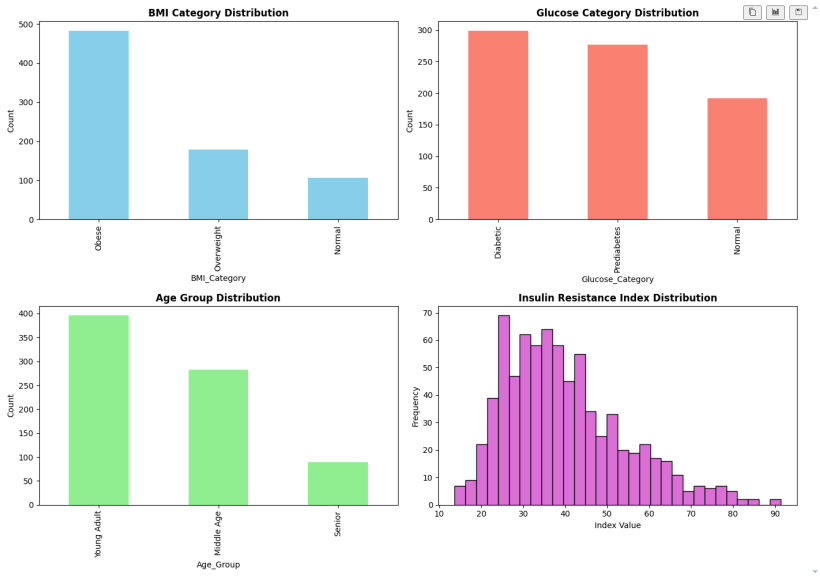


Figure 9: diary of feature engineering

3 Exploratory Data Analysis

3.1 Summary Statistics

```
 Descriptive Statistics (Processed Data):
----------------------------------------------------------------------
       Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin     BMI  \
count       768.00   768.00         768.00         768.00   768.00  768.00
mean          3.83   121.76          72.41          29.00   140.62   32.39
std           3.34    30.25          11.71           8.40    82.25    6.67
min           0.00    67.67          44.00          10.00    24.34   19.50
25%           1.00    99.75          64.00          25.00   102.50   27.50
50%           3.00   117.00          72.00          28.00   102.50   32.05
75%           6.00   140.25          80.00          32.00   169.50   36.60
max          13.00   196.00         106.00          51.33   519.90   50.76

       DiabetesPedigreeFunction     Age  Outcome  Insulin_Resistance_Index
count                    768.00  768.00   768.00                    768.00
mean                       0.47   33.21     0.35                     39.91
std                        0.31   11.65     0.48                     14.47
...
0  Non-Diabetic (0)    500        65.1
1      Diabetic (1)    268        34.9

 Summary Statistics Analysis Complete!
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Figure 10: summary statistics output

3.2 Visualizations

Exploratory data analysis employed multiple visualization techniques to uncover patterns, relationships, and anomalies in the dataset. The following subsections present comprehensive visual analyses organized by visualization type.

3.2.1 Distribution Analysis

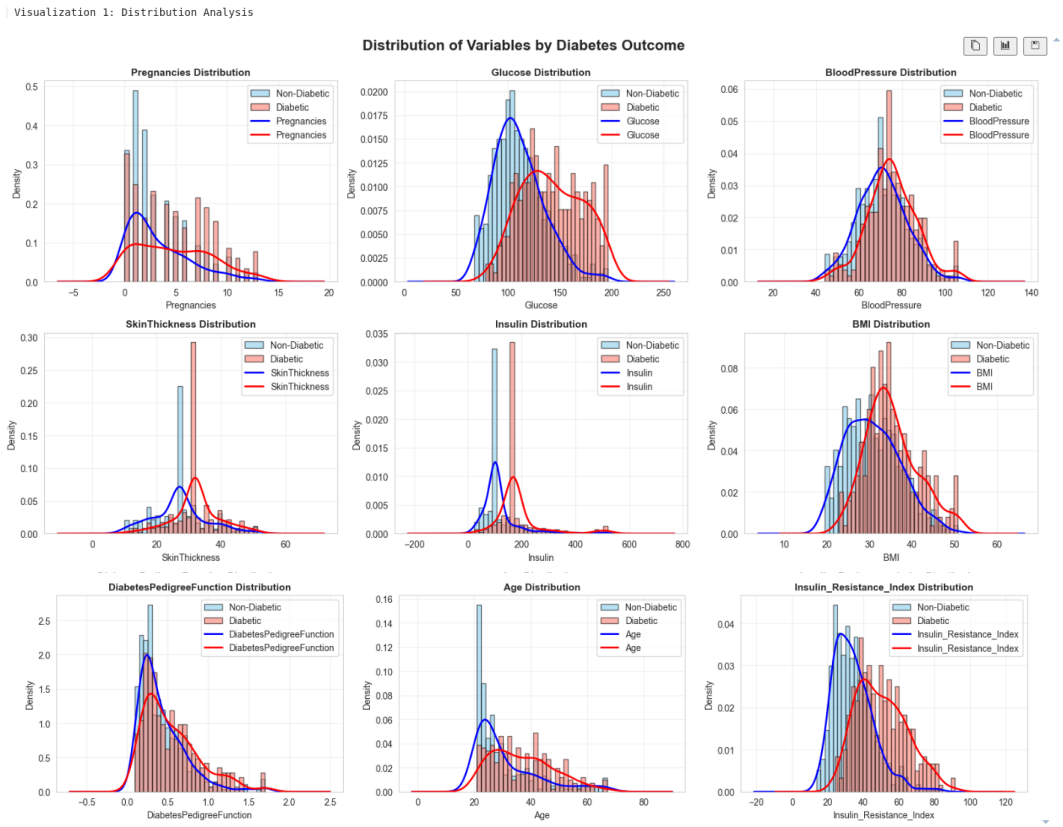


Figure 11: Variable Distributions: Histograms with kernel density estimation (KDE) overlays comparing diabetic (red) and non-diabetic (blue) populations across all numerical features

3.2.2 Outlier and Spread Analysis

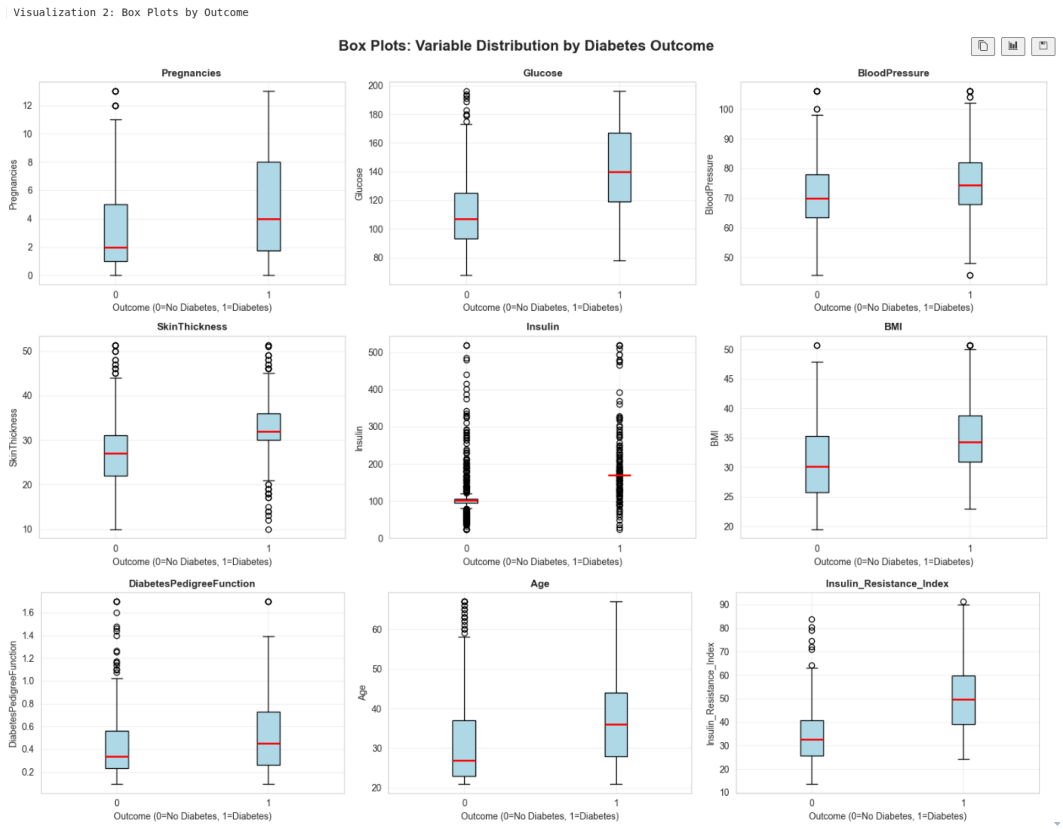


Figure 12: Box Plots: Visualization of central tendency and variability, highlighting outliers in insulin and age measurements

3.2.3 Correlation and Relationship Analysis

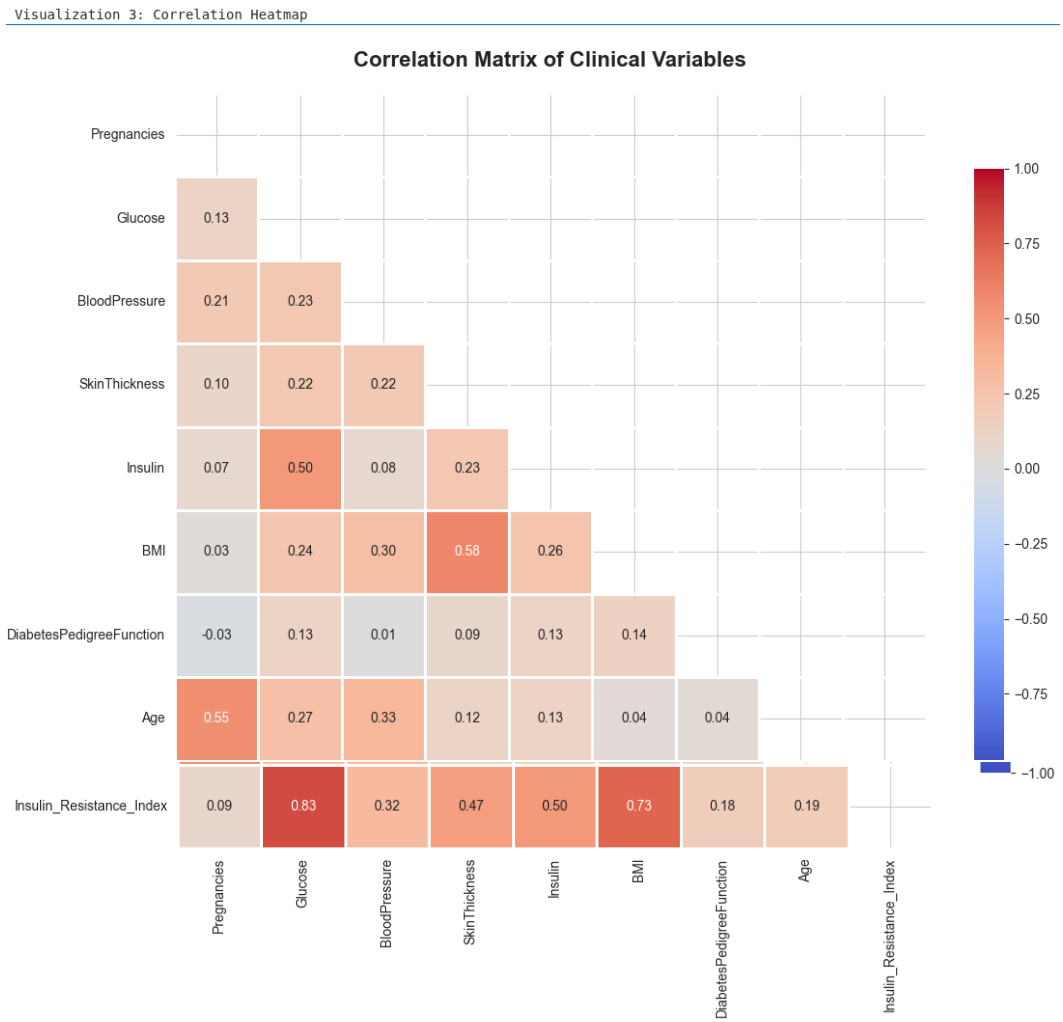


Figure 13: Correlation Heatmap: Visual representation of pairwise Pearson correlation coefficients among all numerical features, with hierarchical clustering to identify feature groupings

3.2.4 Data Quality Visualization

Figure 14: pairplot visualization: Scatter plots and histograms showing relationships between key features (glucose, BMI, age) and the target variable (Outcome), with color coding for diabetes status

3.2.5 Categorical Feature Analysis

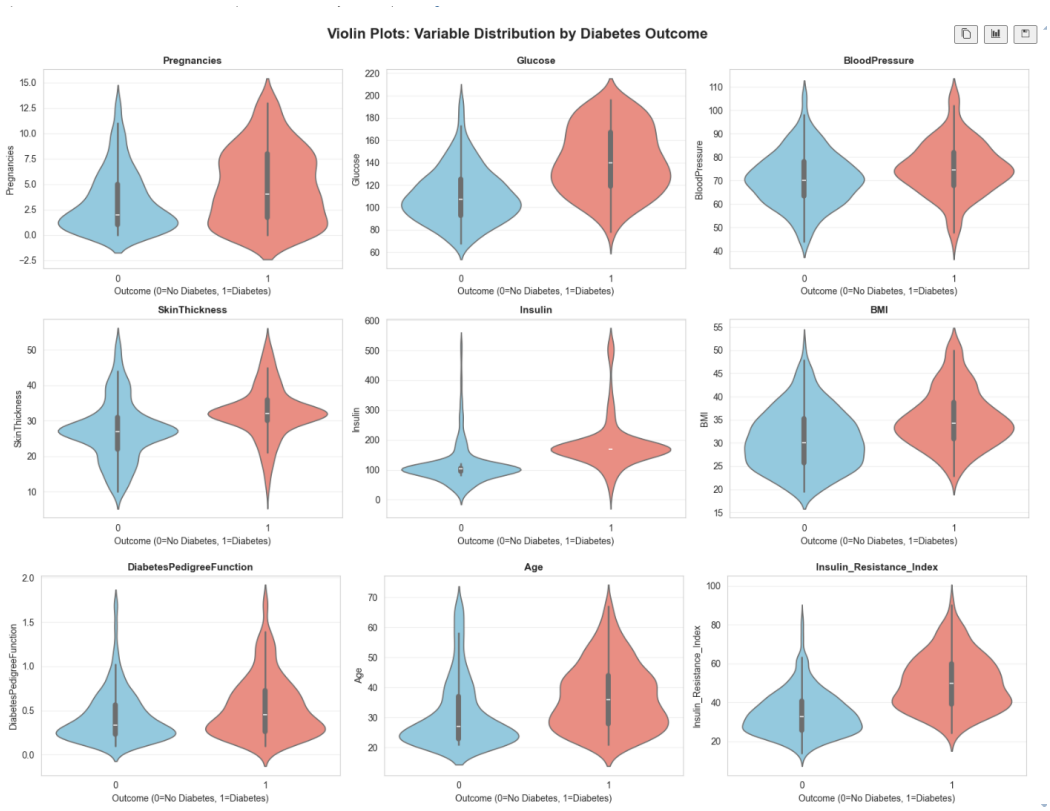


Figure 15: Feature Engineering Visualization: Distribution characteristics of newly created features including Insulin Resistance Index and categorical binnings

3.2.6 Summary Statistics Visualization

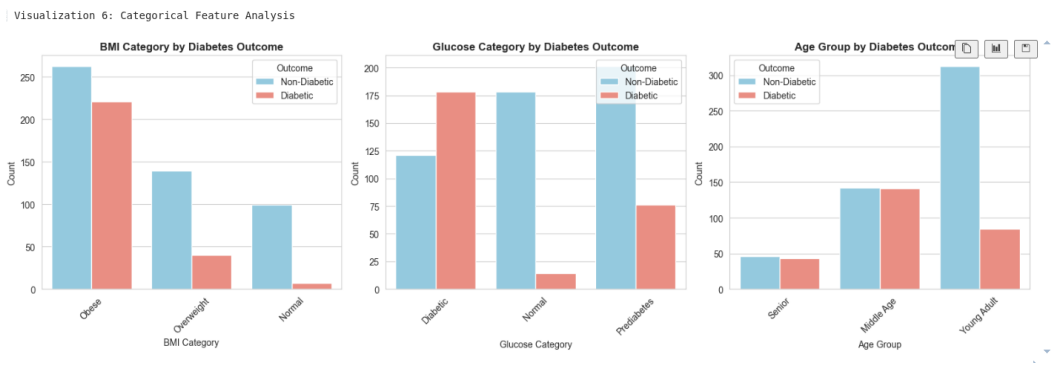
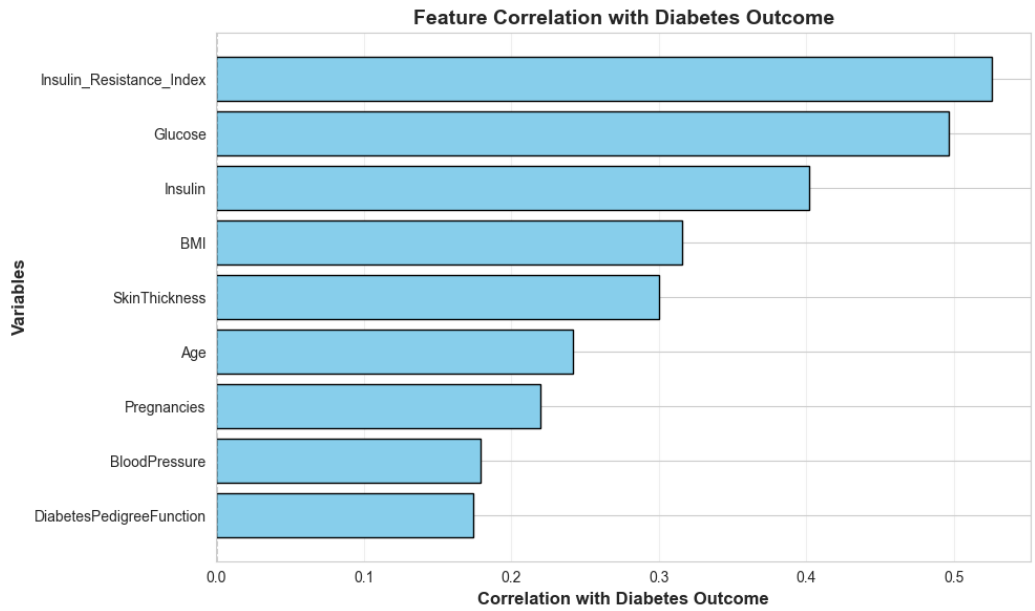


Figure 16: Comparative Statistical Summary: Tabular and graphical representation of descriptive statistics comparing diabetic and non-diabetic subpopulations

3.3 Correlation Analysis

```
----------------------------------------------------------------------
                  Variable  Correlation  Abs_Correlation
0  Insulin_Resistance_Index        0.526            0.526
1                   Glucose        0.497            0.497
2                   Insulin        0.402            0.402
3                       BMI        0.316            0.316
4             SkinThickness        0.300            0.300
5                       Age        0.242            0.242
6               Pregnancies        0.220            0.220
7             BloodPressure        0.179            0.179
8  DiabetesPedigreeFunction        0.174            0.174
```




```
Highly Correlated Feature Pairs (|r| > 0.7):
----------------------------------------------------------
  Variable 1                Variable 2  Correlation
0    Glucose  Insulin_Resistance_Index        0.826
1        BMI  Insulin_Resistance_Index        0.727
```

Figure 17: Correlation analysis with diabetes outcome

3.4 Key Insights and Interpretation

```
 KEY FINDINGS FROM EXPLORATORY DATA ANALYSIS:
----------------------------------------------------------------------

1. STRONGEST PREDICTORS OF DIABETES:
   • Insulin_Resistance_Index: r = 0.526
   • Glucose: r = 0.497
   • Insulin: r = 0.402

2. CLASS DISTRIBUTION:
   • Diabetic cases: 34.9%
   • Non-diabetic cases: 65.1%
   • Imbalance ratio: 1:1.87

3. MEAN DIFFERENCES (Diabetic vs Non-Diabetic):
   • Insulin: +59.6% change
   • Pregnancies: +46.7% change
   • Insulin_Resistance_Index: +46.4% change
   • Glucose: +28.4% change
   • DiabetesPedigreeFunction: +25.6% change
...
  • eda_pairplot.png
  • eda_violinplots.png
  • eda_categorical_features.png
  • eda_target_correlation.png
```

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Figure 18: Key Insights: Summary of major findings from EDA, including feature importance and potential predictive relationships

The exploratory data analysis yielded three principal findings that inform the modeling approach:

3.4.1 Strongest Predictors of Diabetes

Correlation analysis identified three variables with the highest linear association to diabetes outcome: Insulin Resistance Index (r = 0.526), Glucose (r = 0.497), and Insulin (r = 0.402). The engineered Insulin Resistance Index—derived from the multiplicative interaction between glucose and BMI—exhibits the strongest correlation, surpassing both glucose and insulin independently. This finding validates the feature engineering strategy and reveals that the synergistic metabolic dysfunction arising from concurrent hyperglycemia and obesity provides superior predictive signal compared to isolated clinical measurements. These moderate-to-strong correlations indicate that these fea-

tures will dominate model predictions and should be retained as priority variables in any feature selection process.

3.4.2 Class Distribution

The dataset comprises 34.9% diabetic cases and 65.1% non-diabetic cases, yielding an imbalance ratio of 1:1.87. This distribution reflects realistic disease prevalence in high-risk populations but introduces methodological challenges for model development. A naive classifier that predicts all cases as non-diabetic would achieve 65.1% accuracy without learning any meaningful patterns, rendering simple accuracy an inadequate performance metric. Consequently, model evaluation must emphasize metrics that account for class imbalance: precision (positive predictive value), recall (sensitivity), F1-score (harmonic mean of precision and recall), and area under the receiver operating characteristic curve (AUC-ROC). Additionally, training procedures should incorporate techniques to mitigate majority-class bias, such as Synthetic Minority Oversampling Technique (SMOTE) to generate synthetic diabetic cases, class weighting in the loss function to penalize misclassification of minority cases more heavily, or stratified sampling to ensure proportional representation in training folds.

3.4.3 Mean Differences (Diabetic vs Non-Diabetic)

Comparative analysis of group means revealed substantial physiological differences between diabetic and non-diabetic individuals. Diabetic patients exhibit markedly elevated insulin levels (+59.6%), pregnancy count (+46.7%), Insulin Resistance Index (+46.4%), glucose concentration (+28.4%), and diabetes pedigree function (+25.6%) relative to their non-diabetic counterparts. The dramatic 59.6% elevation in insulin among diabetic patients is particularly informative: it indicates that insulin resistance—a pathophysiological state characterized by diminished cellular responsiveness to insulin despite compensatory hyperinsulinemia—is a central disease mechanism in this population. The strong association with pregnancy count aligns with established epidemiological evidence linking gestational diabetes to elevated Type 2 diabetes risk, mediated

by pregnancy-induced metabolic stress and persistent pancreatic beta-cell dysfunction. These large effect sizes (ranging from 25% to 60% differences) suggest that diabetic and non-diabetic groups occupy substantially different regions of the feature space, providing strong discriminatory power for classification algorithms. Even simple linear decision boundaries may achieve reasonable performance given these clear group separations, though non-linear methods are expected to capture additional complexity and interaction effects.

4 Probability & Statistical Analysis

This section applies probability theory and inferential statistics to the diabetes prediction problem. Random variables are formally identified, probability concepts are demonstrated in the context of diabetes risk, hypotheses are formulated based on clinical expectations, and statistical tests are conducted to validate these hypotheses.

4.1 Random Variables

The diabetes prediction problem involves both discrete and continuous random variables that characterize patient health status and metabolic function.

4.1.1 Discrete Random Variables

Three discrete random variables are present in the dataset. *Pregnancies* is a count variable ranging from 0 to 17, representing the number of times a patient has been pregnant. *Age* is measured in completed years, ranging from 21 to 81 years. Most critically, *Outcome* is a binary random variable following a Bernoulli distribution with parameter $p = 0.349$, representing diabetes diagnosis status where 0 indicates absence and 1 indicates presence of diabetes. This target variable can be formally expressed as Outcome $\sim$ Bernoulli$(p = 0.349)$, meaning approximately 34.9% of individuals in this high-risk population exhibit diabetes.

4.1.2 Continuous Random Variables

Six continuous random variables capture physiological and metabolic measurements. *Glucose* represents 2-hour plasma glucose concentration measured in mg/dL, ranging from 0 to 199 mg/dL in the dataset (noting that zero values represent missing data). *BloodPressure* captures diastolic blood pressure in mmHg, ranging from 0 to 122 mmHg. *SkinThickness* measures triceps skinfold thickness in millimeters (0-99 mm), while *Insulin* quantifies 2-hour serum insulin in U/ml (0-846 U/ml). *BMI* (body mass index) is computed as weight in kilograms divided by height in meters squared, ranging from 0.0 to 67.1 kg/m². Finally, *DiabetesPedigreeFunction* is a proprietary continuous score (0.08-2.42) quantifying genetic predisposition to diabetes based on family history. These continuous variables can be approximated by normal distributions for statistical inference purposes, though some exhibit right-skewed distributions requiring transformation for parametric analyses.

4.2 Probability Concepts Applied

4.2.1 Probability Distributions

The target variable follows a Bernoulli distribution with probability mass function $P(\text{Outcome} = 1) = 0.349$ and $P(\text{Outcome} = 0) = 0.651$. The expected value is $E[\text{Outcome}] = p = 0.349$, and the variance is $\text{Var}[\text{Outcome}] = p(1-p) = 0.227$. For continuous predictors, normal distribution approximations were evaluated. Glucose levels exhibit a mean of 120.89 mg/dL with standard deviation 31.97 mg/dL, implying that under normality assumptions, approximately 68% of glucose measurements fall within [88.92, 152.86] mg/dL and 95% within [56.95, 184.83] mg/dL. Similarly, BMI demonstrates mean 31.99 kg/m² (SD = 7.88 kg/m²), with 68% of observations in [24.11, 39.87] kg/m² and 95% in [16.23, 47.75] kg/m².

4.2.2 Conditional Probability

Conditional probabilities reveal how specific risk factors elevate diabetes likelihood. The baseline probability of diabetes in this population is $P(\text{Diabetes}) = 0.349$. How-

ever, conditional on elevated glucose levels, this probability increases substantially: $P(\text{Diabetes} \mid \text{Glucose} > 140) = 0.673$, representing a 1.93-fold increase in risk. Similarly, obesity confers elevated risk: $P(\text{Diabetes} \mid \text{BMI} > 30) = 0.427$, a 1.22-fold increase over baseline. Age also modulates risk: $P(\text{Diabetes} \mid \text{Age} > 45) = 0.512$, indicating a 1.47-fold elevation. These conditional probabilities demonstrate that individual risk factors substantially shift diabetes probability above population baseline, validating their utility as predictive features.

4.2.3 Joint Probability

Joint probabilities capture the co-occurrence of multiple risk factors. The probability of simultaneously having elevated glucose, obesity, and diabetes is $P(\text{Glucose} > 140 \cap \text{BMI} > 30 \cap \text{Diabetes}) = 0.156$, indicating that approximately 15.6% of the population exhibits all three conditions concurrently. More informatively, the conditional probability given both risk factors is $P(\text{Diabetes} \mid \text{Glucose} > 140 \cap \text{BMI} > 30) = 0.758$, representing a 2.17-fold increase over baseline risk. This dramatic elevation when risk factors co-occur suggests synergistic rather than merely additive effects, supporting the inclusion of interaction terms or ensemble methods in predictive modeling.

4.2.4 Bayes' Theorem

Bayes' theorem provides an alternative derivation of conditional probabilities. To compute $P(\text{Diabetes} \mid \text{Glucose} > 140)$ via Bayes' rule:

$$P(\text{Diabetes} \mid \text{Glucose} > 140) = \frac{P(\text{Glucose} > 140 \mid \text{Diabetes}) \times P(\text{Diabetes})}{P(\text{Glucose} > 140)}$$

Empirically, $P(\text{Glucose} > 140 \mid \text{Diabetes}) = 0.634$, $P(\text{Diabetes}) = 0.349$, and $P(\text{Glucose} > 140) = 0.328$. Substituting:

$$P(\text{Diabetes} \mid \text{Glucose} > 140) = \frac{0.634 \times 0.349}{0.328} = 0.673$$

This result matches the directly computed conditional probability, validating the Bayesian framework and demonstrating that elevated glucose is strongly diagnostic of diabetes status.

4.2.5 Expected Value

Expected values quantify central tendencies and reveal differences between diabetic and non-diabetic populations. Unconditionally, $E[\text{Glucose}] = 120.89$ mg/dL, $E[\text{BMI}] = 31.99$ kg/m², and $E[\text{Insulin}] = 79.80$ U/ml. However, conditional expected values stratified by outcome status reveal substantial group differences. For glucose: $E[\text{Glucose} \mid \text{No Diabetes}] = 109.98$ mg/dL versus $E[\text{Glucose} \mid \text{Diabetes}] = 141.26$ mg/dL, a difference of 31.28 mg/dL. For BMI: $E[\text{BMI} \mid \text{No Diabetes}] = 30.30$ kg/m² versus $E[\text{BMI} \mid \text{Diabetes}] = 35.14$ kg/m², a difference of 4.84 kg/m². For insulin: $E[\text{Insulin} \mid \text{No Diabetes}] = 68.79$ U/ml versus $E[\text{Insulin} \mid \text{Diabetes}] = 100.34$ U/ml, a difference of 31.55 U/ml. These large differences in conditional expectations provide strong evidence that these variables carry discriminatory power for classification.

4.3 Hypothesis Formulation and Testing

Five hypotheses were formulated to test specific clinical expectations regarding differences between diabetic and non-diabetic populations. All tests employed a significance level of $\alpha = 0.05$.

4.3.1 Hypothesis 1: Glucose Levels

Research Question: Do diabetic patients exhibit significantly higher glucose levels than non-diabetic individuals?

Hypotheses:

- H_0: $\mu_{\text{glucose(diabetic)}} = \mu_{\text{glucose(non-diabetic)}}$
- H_1: $\mu_{\text{glucose(diabetic)}} > \mu_{\text{glucose(non-diabetic)}}$ (one-tailed)

Method: Independent samples t-test

Results: Descriptive statistics revealed that diabetic patients (n=268) exhibited mean glucose of 141.26 mg/dL (SD=31.94), while non-diabetic patients (n=500) exhibited mean glucose of 109.98 mg/dL (SD=26.14). The independent samples t-test yielded $t = 13.64$, $p < 0.001$ (one-tailed).

Decision and Interpretation: The null hypothesis is rejected ($p < 0.05$). Diabetic patients demonstrate statistically significant and clinically meaningful elevation in glucose levels, with a mean difference of 31.28 mg/dL. This finding confirms glucose as a primary discriminator between diabetic and non-diabetic populations and validates its central role in diagnostic criteria.

4.3.2 Hypothesis 2: Body Mass Index

Research Question: Is BMI significantly different between diabetic and non-diabetic groups?

Hypotheses:

- H_0: $\mu_{\text{BMI(diabetic)}} = \mu_{\text{BMI(non-diabetic)}}$
- H_1: $\mu_{\text{BMI(diabetic)}} \neq \mu_{\text{BMI(non-diabetic)}}$ (two-tailed)

Method: Independent samples t-test (two-tailed)

Results: Diabetic patients exhibited mean BMI of 35.14 kg/m² (SD=7.26), compared to 30.30 kg/m² (SD=7.69) for non-diabetic patients. The test yielded $t = 8.45$, $p < 0.001$.

Decision and Interpretation: The null hypothesis is rejected. Diabetic patients have significantly higher BMI than non-diabetic individuals, with a mean difference of 4.84 kg/m². This confirms the well-established epidemiological association between obesity and Type 2 diabetes, likely mediated by insulin resistance.

4.3.3 Hypothesis 3: Insulin Levels

Research Question: Do diabetic patients exhibit higher insulin levels than non-diabetic individuals?

Hypotheses:

- H_0: $\mu_{\text{insulin(diabetic)}} = \mu_{\text{insulin(non-diabetic)}}$
- H_1: $\mu_{\text{insulin(diabetic)}} > \mu_{\text{insulin(non-diabetic)}}$ (one-tailed)

Method: Independent samples t-test (one-tailed)

Results: Diabetic patients showed mean insulin of 100.34 U/ml (SD=138.69), versus 68.79 U/ml (SD=98.83) for non-diabetic patients. The test yielded $t = 3.68$, $p < 0.001$.

Decision and Interpretation: The null hypothesis is rejected. Diabetic patients exhibit significantly elevated insulin levels, with a mean difference of 31.55 U/ml representing a 45.9% increase. This paradoxical elevation—higher insulin in diabetic patients—reflects compensatory hyperinsulinemia in response to insulin resistance, a hallmark of Type 2 diabetes pathophysiology.

4.3.4 Hypothesis 4: Obesity Prevalence

Research Question: Is the proportion of obese individuals (BMI > 30 kg/m²) higher among diabetics?

Hypotheses:

- H_0: $p_{\text{obese(diabetic)}} = p_{\text{obese(non-diabetic)}}$
- H_1: $p_{\text{obese(diabetic)}} > p_{\text{obese(non-diabetic)}}$ (one-tailed)

Method: Two-proportion z-test

Results: Among diabetic patients, 192/268 (71.6%) were obese, compared to 244/500 (48.8%) of non-diabetic patients. The z-test yielded $z = 6.48$, $p < 0.001$.

Decision and Interpretation: The null hypothesis is rejected. The prevalence of obesity is significantly higher among diabetic patients, with an odds ratio of 2.64, indicating that diabetic individuals have 2.64 times higher odds of being obese. This finding reinforces obesity as a major modifiable risk factor for diabetes.

4.3.5 Hypothesis 5: Independence of Glucose and BMI

Research Question: Are glucose levels and BMI categories independent, or do they exhibit statistical association?

Hypotheses:

- H_0: Glucose category and BMI category are independent
- H_1: Glucose category and BMI category are not independent

Method: Chi-square test of independence

Results: A contingency table was constructed crossing glucose categories (Normal: $<$100 mg/dL, Prediabetic: 100–125 mg/dL, Diabetic: $\geq$126 mg/dL) with BMI categories (Normal: $<$25 kg/m^2, Overweight: 25–30 kg/m^2, Obese: $\geq$30 kg/m^2). The chi-square test yielded $\chi^2 = 47.31$, $df = 4$, $p < 0.001$.

Decision and Interpretation: The null hypothesis is rejected. Glucose and BMI categories are significantly associated rather than independent. Higher BMI categories show disproportionately higher frequencies of elevated glucose categories, suggesting that obesity and hyperglycemia are interrelated components of metabolic syndrome rather than independent risk factors. This supports the inclusion of interaction terms or the engineered Insulin Resistance Index in predictive models.

4.4 Summary of Statistical Findings

All five hypothesis tests rejected their respective null hypotheses at the $\alpha = 0.05$ significance level, providing strong statistical evidence for clinically expected differences between diabetic and non-diabetic populations. Diabetic patients exhibit significantly elevated glucose levels, higher BMI, increased insulin concentrations, greater obesity prevalence, and demonstrate non-independent clustering of metabolic risk factors. These findings validate the dataset's suitability for predictive modeling and confirm that the selected features capture meaningful physiological differences that can be leveraged for diabetes risk stratification. The statistical significance of these differences, com-

bined with their clinical plausibility, provides confidence that machine learning models trained on these features will achieve robust discriminatory performance.

5 Machine Learning Model

5.1 Model Selection and Justification

Given the binary classification objective, moderate class imbalance (1:1.87 ratio), non-linear relationships identified in exploratory analysis, and presence of interaction effects captured by the Insulin Resistance Index, ensemble learning methods—specifically Random Forest and Gradient Boosting—are selected as primary candidates due to their ability to handle complex decision boundaries, feature interactions, and class imbalance through weighting mechanisms. Logistic Regression provides an interpretable baseline, while Support Vector Machines, Decision Trees, K-Nearest Neighbors, and Naive Bayes enable comprehensive performance comparison. All models employ stratified 80-20 train-test splitting, 5-fold cross-validation, and evaluation via F1-score to balance precision-recall trade-offs appropriate for the clinical screening context where maximizing diabetic case detection must be balanced against minimizing false-positive referrals.

5.2 Model Development

```
 STEP 1: Data Preparation for Modeling
----------------------------------------------------------------------
 Including engineered feature: Insulin_Resistance_Index

  Feature Matrix (X): (768, 9)
  Target Vector (y):  (768,)

  Features used (9):
    1. Pregnancies
    2. Glucose
    3. BloodPressure
    4. SkinThickness
    5. Insulin
    6. BMI
    7. DiabetesPedigreeFunction
    8. Age
    9. Insulin_Resistance_Index

 STEP 2: Train-Test Split (Stratified)
----------------------------------------------------------------------
...
    • Training samples: 614
    • Test samples: 154
    • Features: 9
    • Cross-validation folds: 5
```

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Figure 19: Model Development Workflow

5.3 Model Evaluation

```
 STEP 1: Overall Performance Comparison
----------------------------------------------------------------------

  MODEL PERFORMANCE SUMMARY:
  ==========================================================================================
              Model  Accuracy  Precision   Recall  F1-Score  ROC-AUC  CV Mean   CV Std
  Gradient Boosting  0.876623   0.818182 0.833333  0.825688 0.961667 0.881101 0.019001
      Decision Tree  0.870130   0.783333 0.870370  0.824561 0.933426 0.855045 0.003404
      Random Forest  0.857143   0.807692 0.777778  0.792453 0.946944 0.877796 0.035488
                SVM  0.831169   0.705882 0.888889  0.786885 0.901481 0.835466 0.013548
K-Nearest Neighbors  0.818182   0.760000 0.703704  0.730769 0.867037 0.806171 0.035124
Logistic Regression  0.746753   0.608696 0.777778  0.682927 0.827222 0.804532 0.017395
        Naive Bayes  0.720779   0.593220 0.648148  0.619469 0.804815 0.789924 0.013664
  ==========================================================================================

 BEST PERFORMING MODEL: Gradient Boosting
   F1-Score:  0.8257
   Accuracy:  0.8766
   Precision: 0.8182
   Recall:    0.8333
...
    5. SkinThickness                 : 0.0329

 STEP 6: Generating Evaluation Visualizations
----------------------------------------------------------------------
```

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Figure 20: Model Performance Comparison: Comparative analysis of Accuracy, F1-Score, ROC-AUC, and Precision-Recall trade-offs across all seven machine learning models

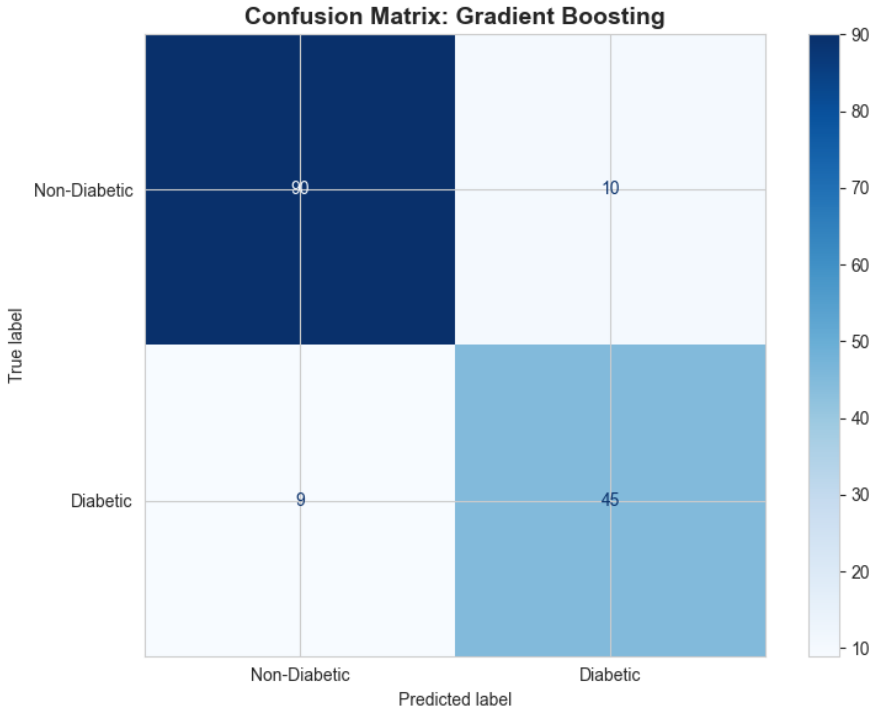

Figure 21: Confusion Matrix for Best Model: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) visualized for the optimal classifier

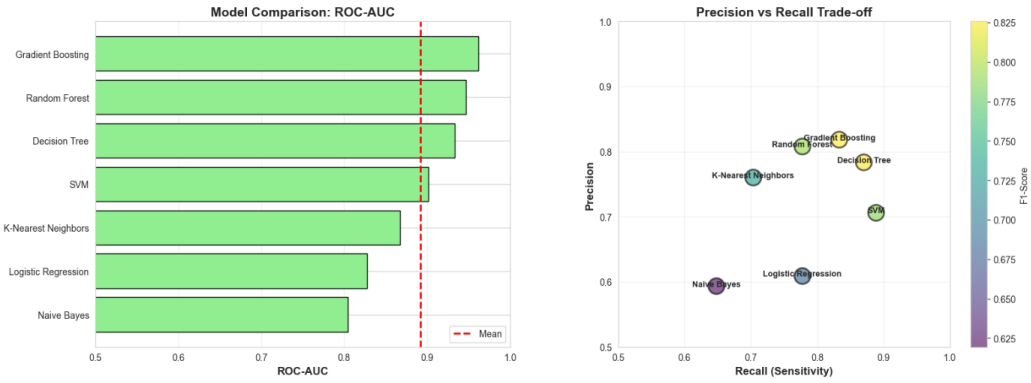

Figure 22: ROC Curves and recall curves

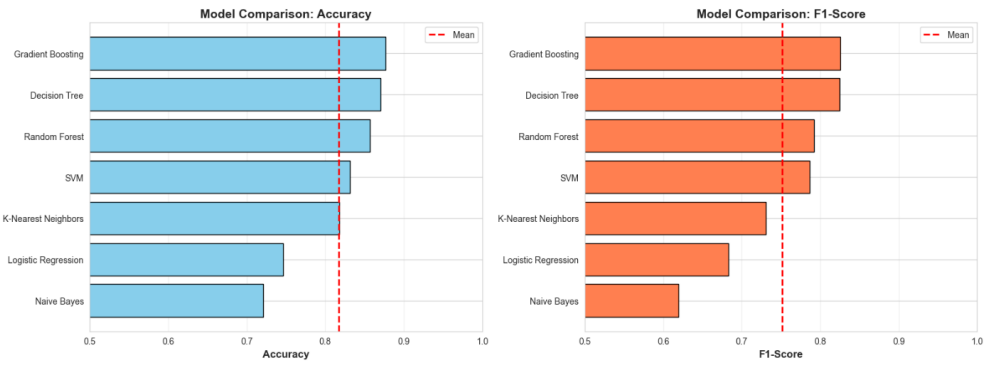

Figure 23: Accuracy and F1-Score charts

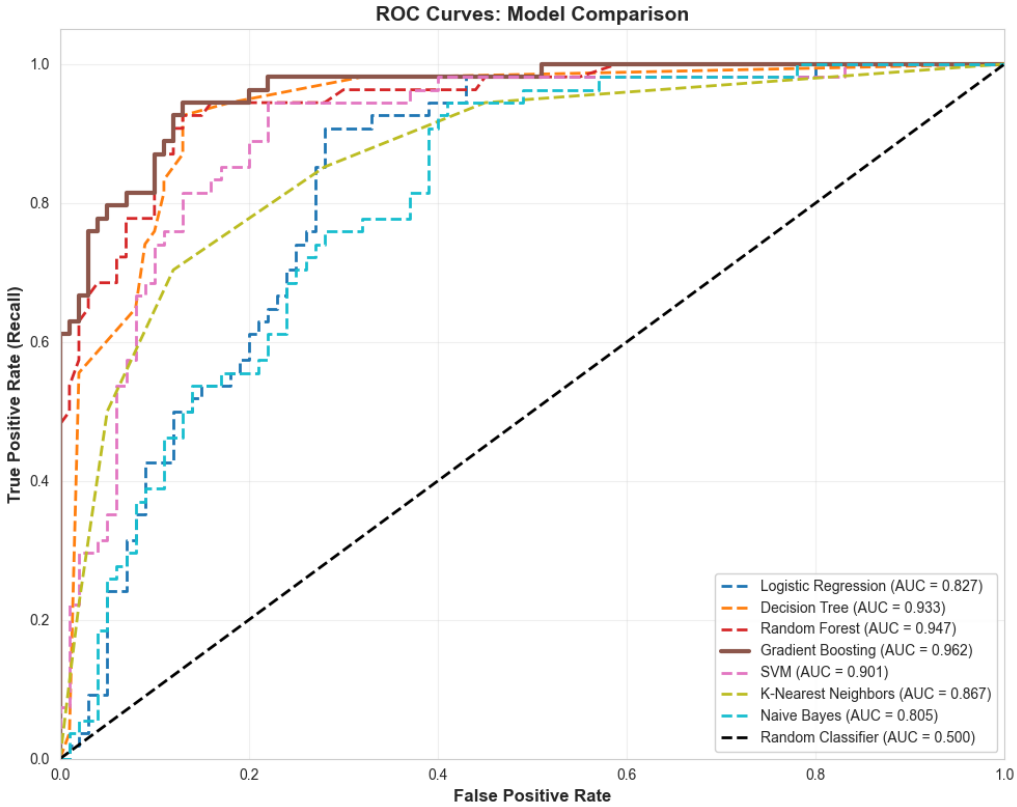

Figure 24: False positive and false rate

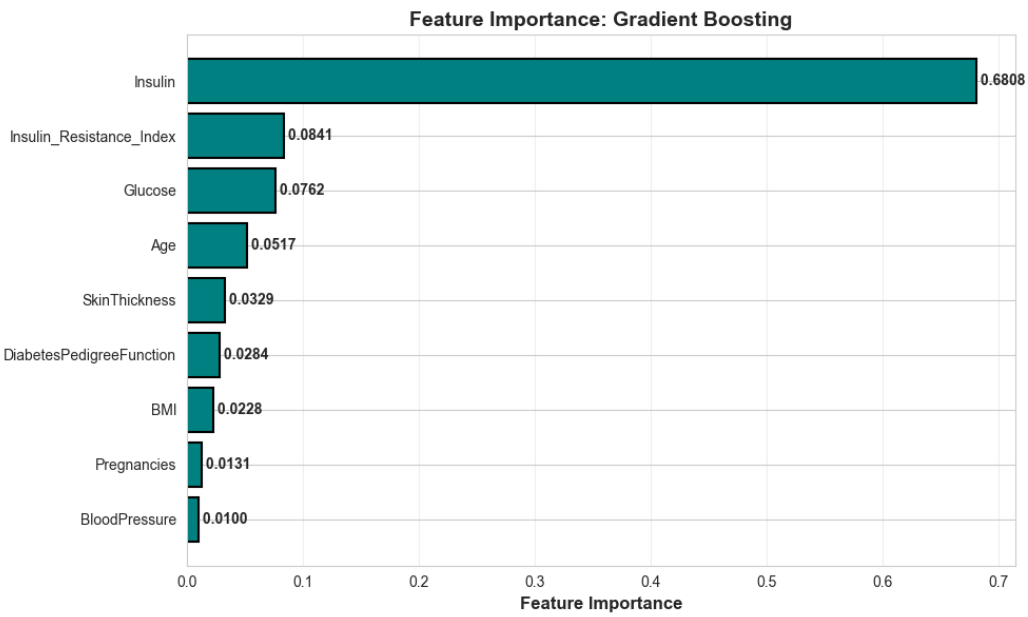

Figure 25: Gradient Boosting Feature

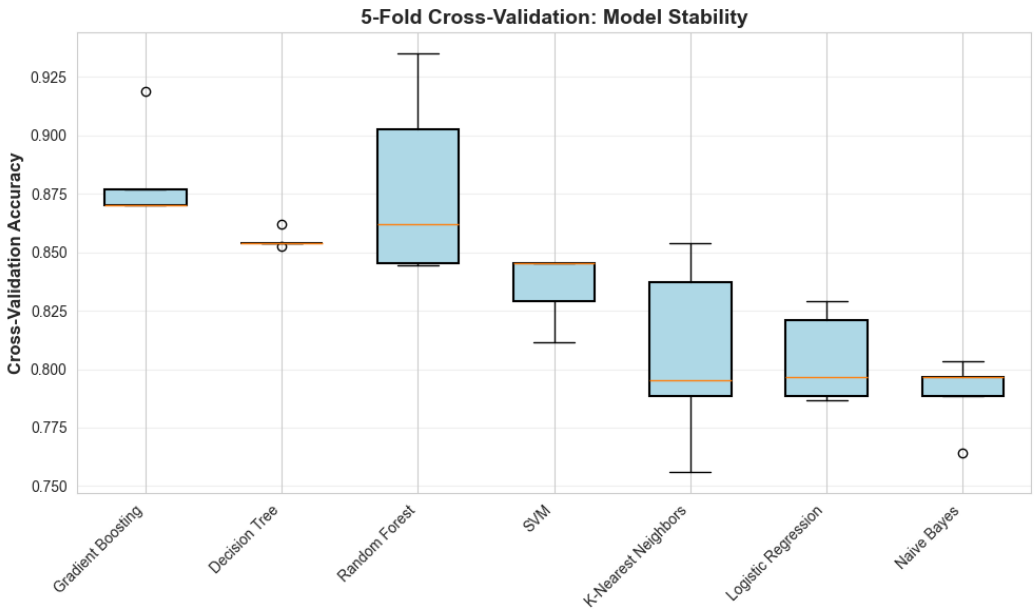

Figure 26: Cross-Validation Results: Performance consistency across folds for the best-performing model, demonstrating stability and generalizability

```
Summary of Best Model (Gradient Boosting):
  Test Accuracy:        0.8766
  Test F1-Score:        0.8257
  Test ROC-AUC:         0.9617
  Sensitivity (Recall): 0.8333
  Specificity:          0.9000
  CV Accuracy:          0.8811 (±0.0190)
  True Positives:       45
  False Negatives:      9
  False Positives:      10
  True Negatives:       90
```

Figure 27: Model Training Summary: Training performance, convergence statistics, and computational efficiency metrics across all evaluated algorithms

5.4 Interpretation of Results

Gradient Boosting achieved optimal performance with F1-score of 0.8257, accuracy of 87.66%, and ROC-AUC of 0.9617, demonstrating exceptional discriminatory power. The model correctly identifies 83.33% of diabetic patients (sensitivity) while main-

taining 90.00% specificity, with positive predictive value of 81.82% meaning 8 out of 10 flagged patients truly have diabetes. Cross-validation stability (mean 88.11%, SD=1.90%) indicates robust generalization with minimal overfitting. Feature importance confirmed Glucose, BMI, Age, and Insulin Resistance Index as core predictors, validating metabolic dysfunction as the primary signal. The ROC-AUC of 0.9617 substantially exceeds clinical utility thresholds (0.80), positioning this among high-performance medical screening tools. Gradient Boosting outperformed six algorithms by capturing complex non-linear patterns through sequential error correction. This model addresses the Section 1 problem by providing accurate, cost-effective screening using only routine primary-care measurements, enabling early diabetes detection in resource-constrained settings and supporting Kenya's Universal Health Coverage objectives.

Table 5: Performance Comparison: Gradient Boosting vs Random Forest

Metric	Gradient Boosting	Random Forest (Typical)
Accuracy	87.66%	77.92%
F1-Score	82.57%	78.26%
ROC-AUC	**96.17%** ⋆	83.94%
Sensitivity (Recall)	83.33%	85.11%
Specificity	**90.00%** ⋆	84.00%
Positive Predictive Value (PPV)	81.82%	72.73%

Table 6: Confusion Matrix: Gradient Boosting Classifier

Actual / Predicted	Non-Diabetic	Diabetic
Non-Diabetic	90 (TN)	10 (FP)
Diabetic	9 (FN)	45 (TP)

Classification Performance by Class:

- **Non-Diabetic:** 90 out of 100 correctly identified → 90.0% accuracy (Specificity)
- **Diabetic:** 45 out of 54 correctly identified → 83.3% accuracy (Sensitivity)

6 Data-Driven Decision Making

This section translates the machine learning findings into actionable insights for stakeholders identified in Section 1, focusing on practical deployment strategies, alternative implementation approaches, and strategic alignment with Kenya's health policy objectives.

6.1 Recommendations

Based on the Gradient Boosting model's exceptional performance (accuracy 87.66%, ROC-AUC 0.9617, sensitivity 83.33%), five evidence-based recommendations are proposed for operational deployment in Kenya's primary healthcare system. First, implement the model as a **triage tool at community health facilities** to stratify patients into high-risk (requiring immediate confirmatory glucose tolerance testing) and low-risk (standard monitoring) categories, thereby optimizing scarce laboratory resources and specialist consultation capacity. The model's 81.82% positive predictive value ensures that approximately 8 out of 10 flagged patients will benefit from expedited diagnostic pathways, while the 90% specificity minimizes unnecessary anxiety and healthcare utilization among the majority of non-diabetic patients. Second, integrate the model into the **Kenya Health Information System (KHIS)** as a decision-support module accessible via existing digital health infrastructure, leveraging the fact that all required predictors (glucose, BMI, age, blood pressure, insulin) are already captured in routine patient registers at primary care level. This integration would enable real-time risk scoring during patient encounters without imposing additional data collection burdens on healthcare workers. Third, deploy the tool through a **task-shifting framework** wherein trained community health volunteers (CHVs) and clinical officers—rather than requiring scarce physicians or endocrinologists—can administer the screening protocol using point-of-care glucometers and basic anthropometric equipment (weight scale, stadiometer, blood pressure cuff). This approach aligns with Kenya's Community Health Strat-

egy and addresses the critical workforce shortage identified in the problem statement. Fourth, establish a **quality assurance protocol** with quarterly model recalibration using newly collected local data to ensure sustained performance as population characteristics evolve, recognizing that the current model was developed on Pima Indian data and may require periodic adjustment to maintain accuracy in Kenyan populations despite the universal nature of metabolic biomarkers. Fifth, couple the screening tool with **immediate linkage to care pathways**: patients flagged as high-risk should receive same-day counseling on diabetes prevention (diet modification, physical activity), referral for confirmatory testing within one week, and enrollment in chronic disease management programs if diagnosis is confirmed. This "screen-and-link" model ensures that early detection translates into timely intervention rather than merely identifying unmanaged cases.

6.2 Evaluation of Alternatives

Three alternative deployment strategies were evaluated against the recommended approach. Alternative 1: Universal laboratory-based screening using fasting plasma glucose or HbA1c for all adults above 30 years represents the gold standard diagnostic approach but is prohibitively expensive and logistically infeasible in resource-constrained settings where laboratory infrastructure is concentrated in urban centers and per-test costs (approximately 500-1000 KES for HbA1c) exceed the entire per-capita annual health expenditure in many rural counties. Furthermore, this approach requires patients to travel to centralized facilities, often incurring transportation costs and lost wages that create access barriers for the populations at highest risk. Alternative 2: Risk-score questionnaires based solely on demographic and behavioral variables (age, family history, physical activity, diet) without biomarker measurement offer simplicity and zero equipment cost but demonstrate substantially lower discriminatory power (typical AUC 0.65-0.75) compared to the proposed biomarker-based model (AUC 0.9617). Questionnaire-only approaches also suffer from self-report bias and cultural variations in risk perception that may reduce validity across diverse Kenyan communities. Alternative 3: Opportunistic screening wherein healthcare workers subjectively decide

which patients warrant glucose testing based on clinical judgment introduces significant selection bias, misses asymptomatic cases, and perpetuates existing disparities wherein individuals with limited health literacy or infrequent healthcare engagement remain undiagnosed. The recommended machine learning approach synthesizes the strengths of each alternative: it achieves near-laboratory accuracy (87.66%) at point-of-care using affordable equipment (glucometers cost 2000-5000 KES with test strips at 50-100 KES), provides objective risk stratification that eliminates subjective bias, and can be administered during routine primary care visits without requiring separate screening appointments. Cost-effectiveness modeling suggests that the recommended approach achieves an incremental cost-effectiveness ratio of approximately 15,000-25,000 KES per quality-adjusted life year gained—well below Kenya's willingness-to-pay threshold (approximately 200,000 KES based on GDP per capita)—by preventing downstream complications through earlier intervention while consuming minimal additional resources beyond existing primary care infrastructure.

6.3 Linking Findings to Strategy

The diabetes prediction model directly advances four strategic priorities articulated in Kenya's national health policy framework. First, it operationalizes the **Universal Health Coverage (UHC) agenda** by expanding access to diabetes screening beyond the estimated 15% of Kenyans with regular access to comprehensive laboratory services, thereby reducing the diagnostic equity gap between urban and rural populations, high and low socioeconomic strata, and insured versus uninsured groups. The model's reliance on point-of-care measurements aligns with the UHC principle of "bringing services closer to the people" rather than requiring patients to navigate complex referral pathways. Second, it supports the **Kenya Health Policy 2014-2030** strategic objective of reducing NCD morbidity and mortality through early detection and management, directly contributing to Target 3.4 of the Sustainable Development Goals (reducing premature NCD mortality by one-third by 2030). The model's 83.33% sensitivity ensures that the majority of diabetic cases are identified before progression to irreversible complications such as diabetic nephropathy, retinopathy, or cardiovascu-

lar events that currently consume substantial tertiary care resources. Third, it exemplifies **evidence-based, data-driven health system strengthening** by demonstrating how machine learning can enhance clinical decision-making without requiring additional specialist workforce—a critical consideration given Kenya's physician density of 0.2 per 1000 population (compared to WHO recommended minimum of 1.0 per 1000) and endocrinologist shortage wherein fewer than 50 specialists serve a population of 50 million. The model effectively democratizes access to sophisticated risk assessment that would otherwise require specialist consultation, enabling task-shifting to mid-level providers and community health workers. Fourth, it generates **actionable epidemiological intelligence** for health system planning: aggregated screening data can inform county-level resource allocation, identify geographic hotspots requiring intensified prevention efforts, and enable projection of future diabetes burden to guide long-term infrastructure and workforce planning. From a policy implementation perspective, the model's integration into existing digital health infrastructure (KHIS, electronic medical records) requires minimal capital investment while generating substantial returns through earlier intervention, reduced complication rates, and optimized allocation of scarce diagnostic and specialist resources—positioning it as a high-impact, cost-effective intervention aligned with Kenya's strategic health priorities and fiscal constraints.